

Syscall catalog (generated)

This table is generated from the syscall numbers in `kernel/include/kernel/syscall.h` and the userspace wrappers in `user/libc/unistd.c`. Calling convention: `int 0x80` with `eax=syscall_no`, args in `ebx/ecx/edx`, return in `eax` (typically `-1` on error).

#	Syscall macro	Userspace wrapper	Register mapping (from wrapper)
1	<code>SYSCALL_WRITE</code>	<code>write(const void *buf, uint32_t len)</code>	<code>eax=SYSCALL_WRITE, ebx=(uint32_t)buf, ecx=len, edx=0</code>
2	<code>SYSCALL_EXIT</code>	<code>``</code>	<code>eax=SYSCALL_EXIT, ebx=?, ecx=?, edx=?</code>
3	<code>SYSCALL_OPEN</code>	<code>open(const char *path)</code>	<code>eax=SYSCALL_OPEN, ebx=(uint32_t)path, ecx=0, edx=0</code>
4	<code>SYSCALL_READ</code>	<code>read(int fd, void *buf, uint32_t len)</code>	<code>eax=SYSCALL_READ, ebx=(uint32_t)fd, ecx=(uint32_t)buf, edx=len</code>
5	<code>SYSCALL_CLOSE</code>	<code>close(int fd)</code>	<code>eax=SYSCALL_CLOSE, ebx=(uint32_t)fd, ecx=0, edx=0</code>
6	<code>SYSCALL_EXEC</code>	<code>exec(const char *path, const char *args, uint32_t args_len)</code>	<code>eax=SYSCALL_EXEC, ebx=(uint32_t)path, ecx=(uint32_t)args, edx=args_len</code>
7	<code>SYSCALL_GETARGS</code>	<code>getargs(char *buf, uint32_t len)</code>	<code>eax=SYSCALL_GETARGS, ebx=(uint32_t)buf, ecx=len, edx=0</code>
8	<code>SYSCALL_STAT</code>	<code>``</code>	<code>eax=SYSCALL_STAT, ebx=?, ecx=?, edx=?</code>
9	<code>SYSCALL_SEEK</code>	<code>lseek(int fd, int offset, int whence)</code>	<code>eax=SYSCALL_SEEK, ebx=(uint32_t)fd, ecx=(uint32_t)offset, edx=(uint32_t)whence</code>
10	<code>SYSCALL_LISTDIR</code>	<code>listdir(const char *path, void *entries, uint32_t max_entries)</code>	<code>eax=SYSCALL_LISTDIR, ebx=(uint32_t)path, ecx=(uint32_t)entries, edx=max_entries</code>
11	<code>SYSCALL_MKDIR</code>	<code>mkdir(const char *path)</code>	<code>eax=SYSCALL_MKDIR, ebx=(uint32_t)path, ecx=0, edx=0</code>
12	<code>SYSCALL_RM</code>	<code>rm(const char *path)</code>	<code>eax=SYSCALL_RM, ebx=(uint32_t)path, ecx=0, edx=0</code>
13	<code>SYSCALL_TOUCH</code>	<code>touch(const char *path)</code>	<code>eax=SYSCALL_TOUCH, ebx=(uint32_t)path, ecx=0, edx=0</code>
14	<code>SYSCALL_GETCWD</code>	<code>getcwd(char *buf, uint32_t len)</code>	<code>eax=SYSCALL_GETCWD, ebx=(uint32_t)buf, ecx=len, edx=0</code>
15	<code>SYSCALL_SETCWD</code>	<code>setcwd(const char *path)</code>	<code>eax=SYSCALL_SETCWD, ebx=(uint32_t)path, ecx=0, edx=0</code>
16	<code>SYSCALL_CLEAR</code>	<code>clear(void)</code>	<code>eax=SYSCALL_CLEAR, ebx=0, ecx=0, edx=0</code>
17	<code>SYSCALL_SETCOLOR</code>	<code>setcolor(uint32_t fg, uint32_t bg)</code>	<code>eax=SYSCALL_SETCOLOR, ebx=fg, ecx=bg, edx=0</code>
18	<code>SYSCALL_WRITEFILE</code>	<code>writefile(const char *path, const void *buf, uint32_t len)</code>	<code>eax=SYSCALL_WRITEFILE, ebx=(uint32_t)path, ecx=(uint32_t)buf, edx=len</code>
19	<code>SYSCALL_HISTORY_COUNT</code>	<code>history_count(void)</code>	<code>eax=SYSCALL_HISTORY_COUNT, ebx=0, ecx=0, edx=0</code>
20	<code>SYSCALL_HISTORY_GET</code>	<code>history_get(uint32_t index, char *buf, uint32_t len)</code>	<code>eax=SYSCALL_HISTORY_GET, ebx=index, ecx=(uint32_t)buf, edx=len</code>
21	<code>SYSCALL_GET_TICKS</code>	<code>get_ticks(void)</code>	<code>eax=SYSCALL_GET_TICKS, ebx=0, ecx=0, edx=0</code>
22	<code>SYSCALL_GET_COMMAND_COUNT</code>	<code>get_command_count(void)</code>	<code>eax=SYSCALL_GET_COMMAND_COUNT, ebx=0, ecx=0, edx=0</code>

23 #	SYSCALL_GETCHAR Syscall macro	getchar(void) Userspace wrapper	eax=SYSCALL_GETCHAR, ebx=0, ecx=0, edx=0 Register mapping (from wrapper)
24	SYSCALL_SLEEP_MS	sleep_ms(uint32_t ms)	eax=SYSCALL_SLEEP_MS, ebx=ms, ecx=0, edx=0
25	SYSCALL_ALIAS_SET	alias_set(const char *name, const char *cmd)	eax=SYSCALL_ALIAS_SET, ebx=(uint32_t)name, ecx=(uint32_t)cmd, edx=0
26	SYSCALL_ALIAS_REMOVE	alias_remove(const char *name)	eax=SYSCALL_ALIAS_REMOVE, ebx=(uint32_t)name, ecx=0, edx=0
27	SYSCALL_ALIAS_COUNT	alias_count(void)	eax=SYSCALL_ALIAS_COUNT, ebx=0, ecx=0, edx=0
28	SYSCALL_ALIAS_GET	alias_get(uint32_t index, char *name, char *cmd)	eax=SYSCALL_ALIAS_GET, ebx=index, ecx=(uint32_t)name, edx=(uint32_t)cmd
29	SYSCALL_TIMER_START	timer_start(void)	eax=SYSCALL_TIMER_START, ebx=0, ecx=0, edx=0
30	SYSCALL_TIMER_STOP	timer_stop(void)	eax=SYSCALL_TIMER_STOP, ebx=0, ecx=0, edx=0
31	SYSCALL_TIMER_STATUS	timer_status(void)	eax=SYSCALL_TIMER_STATUS, ebx=0, ecx=0, edx=0
32	SYSCALL_BEEP	beep(uint32_t frequency_hz, uint32_t duration_ms)	eax=SYSCALL_BEEP, ebx=frequency_hz, ecx=duration_ms, edx=0
33	SYSCALL_HALT	halt(void)	eax=SYSCALL_HALT, ebx=0, ecx=0, edx=0
34	SYSCALL_GFX_DEMO	gfx_demo(void)	eax=SYSCALL_GFX_DEMO, ebx=0, ecx=0, edx=0
35	SYSCALL_GFX_ANIM	gfx_anim(void)	eax=SYSCALL_GFX_ANIM, ebx=0, ecx=0, edx=0
36	SYSCALL_GFX_PAINT	gfx_paint(const char *path)	eax=SYSCALL_GFX_PAINT, ebx=(uint32_t)path, ecx=0, edx=0
37	SYSCALL_GUI_DESKTOP	gui_desktop(void)	eax=SYSCALL_GUI_DESKTOP, ebx=0, ecx=0, edx=0
38	SYSCALL_GUI	gui_run(void)	eax=SYSCALL_GUI, ebx=0, ecx=0, edx=0
39	SYSCALL_GUI_PAINT	gui_paint(const char *path)	eax=SYSCALL_GUI_PAINT, ebx=(uint32_t)path, ecx=0, edx=0
40	SYSCALL_GUI_CALC	gui_calc(void)	eax=SYSCALL_GUI_CALC, ebx=0, ecx=0, edx=0
41	SYSCALL_GUI_FILEMGR	gui_filemgr(void)	eax=SYSCALL_GUI_FILEMGR, ebx=0, ecx=0, edx=0
42	SYSCALL_GFX_SET_MODE	``	eax=SYSCALL_GFX_SET_MODE, ebx=?, ecx=?, edx=?
43	SYSCALL_GFX_GET_MODE	``	eax=SYSCALL_GFX_GET_MODE, ebx=?, ecx=?, edx=?
44	SYSCALL_GFX_GET_WIDTH	``	eax=SYSCALL_GFX_GET_WIDTH, ebx=?, ecx=?, edx=?
45	SYSCALL_GFX_GET_HEIGHT	``	eax=SYSCALL_GFX_GET_HEIGHT, ebx=?, ecx=?, edx=?
46	SYSCALL_GFX_CLEAR	``	eax=SYSCALL_GFX_CLEAR, ebx=?, ecx=?, edx=?
47	SYSCALL_GFX_PUTPIXEL	``	eax=SYSCALL_GFX_PUTPIXEL, ebx=?, ecx=?, edx=?
48	SYSCALL_GFX_DRAW_RECT	``	eax=SYSCALL_GFX_DRAW_RECT, ebx=?, ecx=?, edx=?
49	SYSCALL_GFX_FILL_RECT	``	eax=SYSCALL_GFX_FILL_RECT, ebx=?, ecx=?, edx=?
50	SYSCALL_GFX_DRAW_LINE	``	eax=SYSCALL_GFX_DRAW_LINE, ebx=?, ecx=?, edx=?
51	SYSCALL_GFX_DRAW_CHAR	``	eax=SYSCALL_GFX_DRAW_CHAR, ebx=?, ecx=?, edx=?
52	SYSCALL_GFX_PRINT	``	eax=SYSCALL_GFX_PRINT, ebx=?, ecx=?, edx=?

53	SYSCALL_GFX_FLIP Syscall macro	`` Userspace wrapper	eax=SYSCALL_GFX_FLIP, ebx=?, ecx=?, edx=? Register mapping (from wrapper)
54	SYSCALL_GFX_DOUBLEBUFFER_ENABLE	``	eax=SYSCALL_GFX_DOUBLEBUFFER_ENABLE, ebx=?, ecx=?, edx=?
55	SYSCALL_GFX_DOUBLEBUFFER_DISABLE	``	eax=SYSCALL_GFX_DOUBLEBUFFER_DISABLE, ebx=?, ecx=?, edx=?
56	SYSCALL_MOUSE_GET_STATE	``	eax=SYSCALL_MOUSE_GET_STATE, ebx=?, ecx=?, edx=?
57	SYSCALL_KEYBOARD_HAS_INPUT	keyboard_has_input(void)	eax=SYSCALL_KEYBOARD_HAS_INPUT, ebx=0, ecx=0, edx=0
58	SYSCALL_RENAME	rename(const char *old_path, const char *new_name)	eax=SYSCALL_RENAME, ebx=(uint32_t)old_path, ecx=(uint32_t)new_name, edx=0
59	SYSCALL_SPAWN	spawn(const char *path, const char *args, uint32_t args_len)	eax=SYSCALL_SPAWN, ebx=(uint32_t)path, ecx=(uint32_t)args, edx=args_len
60	SYSCALL_WAIT	wait(int *status)	eax=SYSCALL_WAIT, ebx=(uint32_t)-1, ecx=(uint32_t)status, edx=0
61	SYSCALL_FORK	fork(void)	eax=SYSCALL_FORK, ebx=0, ecx=0, edx=0
62	SYSCALL_GFX_BLIT	``	eax=SYSCALL_GFX_BLIT, ebx=?, ecx=?, edx=?
63	SYSCALL_KEY_REPEAT	keyboard_set_repeat(uint8_t delay, uint8_t rate)	eax=SYSCALL_KEY_REPEAT, ebx=(uint32_t)delay, ecx=(uint32_t)rate, edx=0
64	SYSCALL_SPEAKER_START	speaker_start(uint32_t frequency_hz)	eax=SYSCALL_SPEAKER_START, ebx=frequency_hz, ecx=0, edx=0
65	SYSCALL_SPEAKER_STOP	speaker_stop(void)	eax=SYSCALL_SPEAKER_STOP, ebx=0, ecx=0, edx=0
66	SYSCALL_FS_FREE_BLOCKS	fs_get_free_blocks(void)	eax=SYSCALL_FS_FREE_BLOCKS, ebx=0, ecx=0, edx=0
67	SYSCALL_HEAP_STATS	heap_get_stats(user_heap_stats_t *stats)	eax=SYSCALL_HEAP_STATS, ebx=(uint32_t)stats, ecx=sizeof(*stats), edx=0
68	SYSCALL_PROCESS_COUNT	process_count(void)	eax=SYSCALL_PROCESS_COUNT, ebx=0, ecx=0, edx=0
69	SYSCALL_PROCESS_LIST	process_list(user_process_info_t *out, uint32_t max_entries)	eax=SYSCALL_PROCESS_LIST, ebx=(uint32_t)out, ecx=max_entries, edx=0
70	SYSCALL_INSTALL_EMBEDDED	install_embedded(const char *path)	eax=SYSCALL_INSTALL_EMBEDDED, ebx=(uint32_t)path, ecx=0, edx=0
71	SYSCALL_BRK	brk(void *addr)	eax=SYSCALL_BRK, ebx=(uint32_t)addr, ecx=0, edx=0
72	SYSCALL_PIPE	pipe(int fds[2])	eax=SYSCALL_PIPE, ebx=(uint32_t)fds, ecx=0, edx=0
73	SYSCALL_DUP2	dup2(int oldfd, int newfd)	eax=SYSCALL_DUP2, ebx=(uint32_t)oldfd, ecx=(uint32_t)newfd, edx=0
74	SYSCALL_KILL	kill(int pid, int sig)	eax=SYSCALL_KILL, ebx=(uint32_t)pid, ecx=(uint32_t)sig, edx=0
75	SYSCALL_AUDIO_WRITE	audio_write(const void *buf, uint32_t bytes)	eax=SYSCALL_AUDIO_WRITE, ebx=(uint32_t)buf, ecx=bytes, edx=0
76	SYSCALL_AUDIO_SET_VOLUME	audio_set_volume(uint8_t master, uint8_t pcm)	eax=SYSCALL_AUDIO_SET_VOLUME, ebx=master, ecx=pcm, edx=0
77	SYSCALL_AUDIO_GET_VOLUME	audio_get_volume(uint8_t *master, uint8_t *pcm)	eax=SYSCALL_AUDIO_GET_VOLUME, ebx=0, ecx=0, edx=0

#	syscall macro	Userspace wrapper	Register mapping (from wrapper)
78	<code>SYSCALL_AUDIO_STATUS</code>	<code>audio_is_ready(void)</code>	<code>eax=SYSCALL_AUDIO_STATUS, ebx=0, ecx=0, edx=0</code>